

Erell Adventures!

From the Field: 2024 Highlights and
What's Next



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Meet the Founders of Erell

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Doug

PAST RESEARCH:

How do lizards avoid becoming lunch? Erell researchers tackled this question with studies spanning continents:

- In Texas, *Aspidoscelis inornatus* employs speed and camouflage to evade predators.
- In Namibia, *Meroles* species use zig-zagging movements to outwit threats in sandy dunes.
- On Dominica, *Pholidoscelis fuscatus* showcases unique hiding strategies amidst lush island terrain.

Dr. Douglas Eifler is a passionate ecologist, herpetologist, and the Chief Investigator at Erell Institute. With a PhD in Organismic and Evolutionary Biology from Harvard and a background spanning wildlife ecology, philosophy, and mathematics, Doug's curiosity knows no bounds. Whether tracking lizards or exploring new scientific frontiers, Doug brings enthusiasm and innovation to everything he does.



Maria

Dr. Maria Eifler is the Chief of Operations and Organization at Erell Institute. With a PhD in Systematics and Ecology from KU, her expertise spans behavioral & wildlife ecology, conservation biology, mammalogy, and natural history. She was recognized with the KU Undergraduate Research Mentor Award in 2017 and continues to inspire through her teaching, research, and extensive scientific endeavors and publications.

Italian Wall Lizards

Podarcis siculus is particularly notable for its ability to adapt to a wide range of urban and suburban environments. In the United States, established populations exist not only in Kansas but also in major cities such as New York City, Philadelphia, and Los Angeles, highlighting an impressive ecological range that extends beyond its native Mediterranean climate. Our research seeks to understand how urban environments influence the species physiologically and behaviorally. How do introduced populations differ from their native counterparts? And within Douglas County, what types of habitats do they prefer?

RESEARCH INITIATIVES AND GOALS

- ✓ Identify the key traits that enable Italian wall lizards to successfully invade diverse habitats.
- ✓ Determine the physiological, behavioral, and ecological factors that facilitate their spread.
- ✓ Assess their potential role in altering local biodiversity and ecosystem dynamics.

Want to know more about Italian Wall Lizard Research & how to become involved? Visit <https://www.erell.ngo/research/italian-wall-lizards>

Lizard Camp: Building the Next Generation of Scientists

Lizard Camp is a transformative hands-on experience designed to inspire and train students from underrepresented backgrounds in science. Each year, teams of undergraduate, post-baccalaureate, and graduate students work collaboratively to study lizards in desert ecosystems worldwide.

Recent Lizard Camp Destinations:

2024 (Chihuahuan Desert, TX): Eight students from three universities studied natural history and lizard behavior.

2023 (Bouhedma National Park, Tunisia): A cross-cultural team of 8 Tunisian students, 2 U.S. assistants, and 2 Tunisian collaborators explored how morphology influences habitat use.

Errell Trips and Events of 2024

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Mongolia

This summer, Doug and Maria, alongside two U.S. students, partnered with Dr. Bazartseren Boldgiv and his lab at the National University of Mongolia for a research expedition in the Gobi Desert. For three weeks, the team camped near Sainshand, Dornogovi, studying the movement patterns of adult male, adult female, and juvenile Tuvan toad-headed agamas (*Phrynocephalus versicolor*).



Movement Workshop

We held a 1-day workshop focused on the principles and technologies inherent in movement ecology in Lawrence, Kansas on 2 November 2024. There were 16 attendees coming from 5 different institutions, including advanced high school students, undergraduates, and professional biologists. The morning consisted of classroom presentations and activities, and then we spent the afternoon engaged in GPS field exercises.



Dalquest Research Station, TX

Our time in the field focused on honing essential techniques for tracking lizards, including mapping their movement patterns & measuring distances traveled.



Beyond research, we explored the rugged desert landscape, hiked through diverse habitats, and gained hands-on experience in fieldwork methods critical for studying lizard behavior and movement.



SEEDS

Spreading Opportunity and Knowledge

In October 2024, Erell received official approval to establish a regional SEEDS (Strategies for Ecology, Education, Diversity, and Sustainability) chapter through the Ecological Society of America. The club was founded to connect high school, undergraduate, and graduate students across Kansas and beyond, offering hands-on opportunities for outdoor ecological research and education.

In just a few months since its recognition, Erell SEEDS has expanded to include students from four campuses. Our chapter has already begun meaningful engagement, including sending a student to a SEEDS regional field trip at the University of Virginia Coastal Research Center in May. Currently, we're developing a grant proposal to ESA to support a fall field trip exploring the ecological and cultural history of the Kansas River.

What's Next?

This fall, one of our members, Dahtiya Stanley, will attend the 2025 Ecological Society of America's SEEDS Leadership Meeting from September 24-28 at Black Rock Forest in Cornwall, New York. The event's theme is "Carbon Credits - The Future of an Ecology Economy," led by ESA President Dr. Peter Groffman in collaboration with the Cary Institute of Ecosystem Studies. Participants will work together to develop new programming that benefits the wider SEEDS community and helps shape the future of ecological education and leadership.



**Ecology,
Education,
Diversity,
Sustainability (&
Friendships!!)**

Projects of 2025



Lizard Camp 2025

A month-long training program at the Dalquest Desert Research Station took place this June. Throughout the month, students engaged in every stage of the scientific process: project design, data collection, analysis, and presentation. The camp also included natural history discussions, computer workshops, off-site field trips, and daily hikes, all while building essential fieldwork skills such as behavioral observation, habitat measurement, and ecological data recording.



Namibia Expedition

At the beginning of 2025, Doug and Maria undertook an adventure in Southern Africa, scoping out possibilities for new research on lizard movement and predator-prey interactions of Desert Monitor Lizards.

"We were able to connect with two of our former students who are now wildlife biology professionals in Namibia, and created new contacts with Namibian National Park biologists, all of whom were very excited about future research collaborations. We anticipate undertaking a project centered in northern Namibia, during September 2026, when monitor lizards are most active." — Maria

Future Projects of 2025 and Beyond



Opportunities in Paraguay...

In April 2025, Doug and María embarked on another month-long expedition to Paraguay, surveying additional sites for future research. Looking ahead, our team aims to build expertise in advanced telemetry technologies, which will play a key role in full-scale ecological studies at the locations they scouted.

"Our pilot expedition to Paraguay was very successful! We visited 3 different habitats in distant parts of the country (Atlantic Forest in the east, wetlands in the southwest, and Chaco in the west-central area of Paraguay), and identified potential collaborators at each site. We are most interested in initiating a project on tegu lizard movement ecology in the northwest part of Paraguay, deeper into the Chaco than we were able to visit, as recent rains had washed out the roads! We plan to return in a few months (September 2025) when conditions are drier and the lizards are more active to deploy GPS radio transmitters and initiate the project."



Baja California Sur

Looking ahead, we are building international partnerships to expand our field experiences. Our colleagues in the Behavior Lab at the Autonomous University of the State of Mexico (UAEM), led by Dra. María de Lourdes Ruiz-Gómez, have been exploring sites in Baja California Sur as potential venues for collaborative research. We are currently working toward organizing a short field camp in Baja with their team, anticipated for spring 2026.